

**Sheth L.U.J & Sir M.V College Of Science**  
**Subject :- Data visualization with Power BI & Tableau**  
**Practical No 11**

**Aim :- Association Rule Mining**

- **Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.**
- **Generate association rules from the frequent itemsets**

**Description :-**

The code loads the Groceries dataset and groups products into individual customer transactions. It uses the Apriori algorithm to identify frequent itemsets based on minimum support. Association rules are then generated using confidence and lift to find relationships between products. Finally, the code displays the top frequent itemsets and association rules using bar graphs.

**Code :-**

```
T092 practical no 9.py - C:\Users\Vinod Mali\Desktop\T092 practical no 9.py (3,14.6)
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules

file_path = "Groceries_dataset.csv"

df = pd.read_csv(file_path)

print("Dataset loaded successfully!")
print("\nFirst 5 rows:")
print(df.head())

print("\nDataset shape:")
print(df.shape)

print("\nColumn names:")
print(df.columns.tolist())

transactions = df.groupby(
    ["Member_number", "Date"]
)[["itemDescription"]].apply(list).values.tolist()

print("\nNumber of transactions:", len(transactions))

te = TransactionEncoder()

te_array = te.fit(transactions).transform(transactions)
```

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```
T092: practical no 9.py - C:/Users/Vinod Mali/Desktop/AI/T092: practical no 9.py (3,14.6)
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basket = pd.DataFrame(
    te_array,
    columns=te.columns_
)

print("\nTransaction Matrix:")
print(basket.head())

frequent_itemsets = apriori(
    basket,
    min_support=0.01,
    use_colnames=True
)

frequent_itemsets['length'] = frequent_itemsets['itemsets'].apply(len)

print("\nFrequent Itemsets:")
print(frequent_itemsets.sort_values(
    by='support',
    ascending=False
).head(20))

rules = association_rules(
    frequent_itemsets,
    metric="confidence",
    min_threshold=0.30
)

rules = rules[
    [
        'antecedents',
        'consequents',
        'support',
        'confidence',
        'lift'
    ]
]

rules = rules.sort_values(
    by='lift',
    ascending=False
)

print("\nAssociation Rules:")
print(rules.head(20).to_string(index=False))

strong_rules = rules[
    (rules['confidence'] >= 0.50) &
    (rules['lift'] > 1)
]

print("\nStrong Association Rules:")
print(strong_rules.head(20).to_string(index=False))
```

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```
T092: practical no 9.py - C:/Users/Vinod Mali/Desktop/AI/T092: practical no 9.py (3,14,6)
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top_itemsets = frequent_itemsets.sort_values(
    by='support',
    ascending=False
).head(10)

itemset_labels = [
    ', '.join(list(itemset))
    for itemset in top_itemsets['itemsets']
]

plt.figure(figsize=(10, 6))

plt.barh(
    itemset_labels[:-1],
    top_itemsets['support'].values[:-1]
)

plt.xlabel("Support")
plt.ylabel("Itemset")
plt.title("Top 10 Frequent Itemsets")

plt.tight_layout()
plt.show()

top_rules = rules.head(10).copy()

rule_labels = [
    ', '.join(list(row['antecedents'])) +
    ' -> ' +
    ', '.join(list(row['consequents']))
    for _, row in top_rules.iterrows()
]

plt.figure(figsize=(10, 6))

plt.barh(
    rule_labels[:-1],
    top_rules['lift'].values[:-1]
)

plt.xlabel("Lift")
plt.ylabel("Association Rule")
plt.title("Top 10 Association Rules by Lift")

plt.tight_layout()
plt.show()

print("\n=====")
print("ASSOCIATION RULE MINING SUMMARY")
print("=====")

print("Total transactions:", len(transactions))

print(
    "Number of frequent itemsets:",
    len(frequent_itemsets)
)
```

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```
print("Total transactions:", len(transactions))

print(
    "Number of frequent itemsets:",
    len(frequent_itemsets)
)

print(
    "Number of association rules:",
    len(rules)
)

print(
    "Number of strong rules:",
    len(strong_rules)
)

print("\nTop 5 Rules:")
print(rules.head(5).to_string(index=False))
```

**OutPut :-**

```
"IDLE Shell 3.14.6"
File Edit Shell Debug Options Window Help

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/Vinod Mali/Desktop/Ai/T092 practical no 9.py =====
Dataset loaded successfully!

First 5 rows:
  Member_number  Date  itemDescription
0         1808  21-07-2015  tropical fruit
1         2552  05-01-2015    whole milk
2         2300  19-09-2015    pip fruit
3         1187  12-12-2015  other vegetables
4         3037  01-02-2015    whole milk

Dataset shape:
(38765, 3)

Column names:
['Member_number', 'Date', 'itemDescription']

Number of transactions: 14963

Transaction Matrix:
Instant food products  UHT-milk  ...  yogurt  zwieback
0          False  False  ...    True    False
1          False  False  ...   False    False
2          False  False  ...   False    False
3          False  False  ...   False    False
4          False  False  ...   False    False
```

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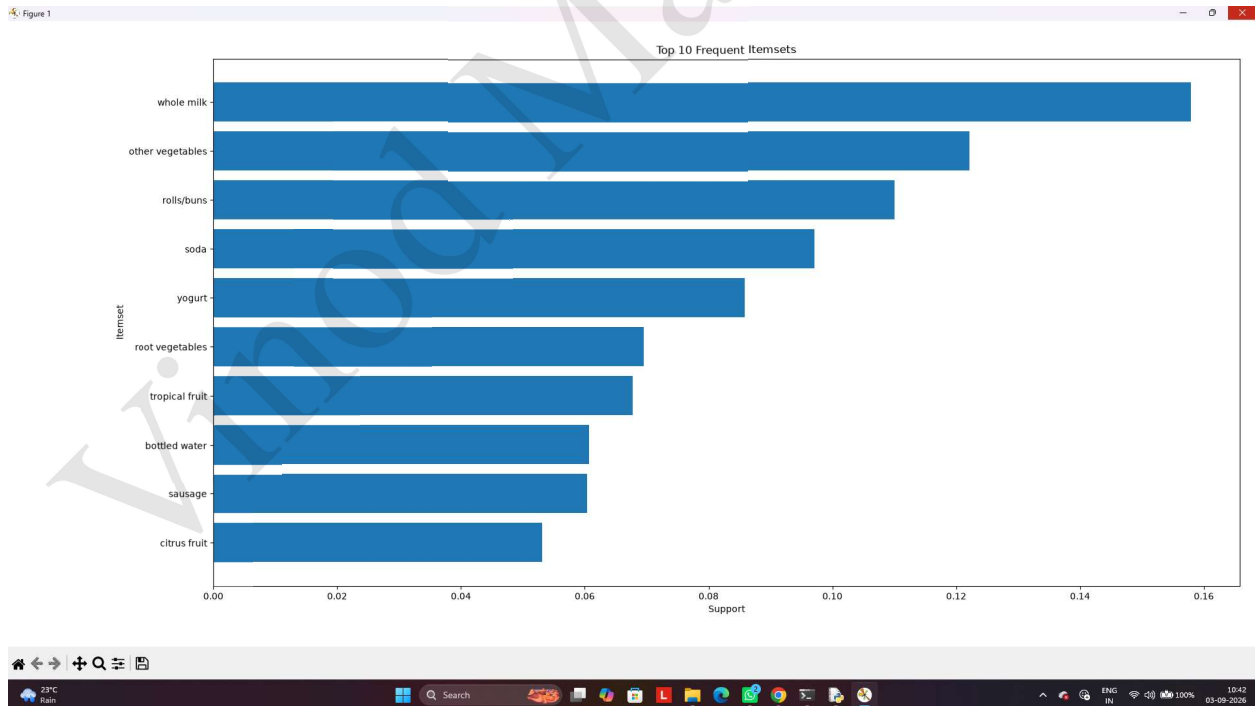
```
"IDLE Shell 3.14.0"
File Edit Shell Debug Options Window Help

[5 rows x 167 columns]

Frequent Itemsets:
support      itemsets length
62 0.157923   frozenset({whole milk}) 1
40 0.122101   frozenset({other vegetables}) 1
46 0.110005   frozenset({rolls/buns}) 1
52 0.097106   frozenset({soda}) 1
63 0.085879   frozenset({yogurt}) 1
47 0.069572   frozenset({root vegetables}) 1
57 0.067767   frozenset({tropical fruit}) 1
5 0.060683    frozenset({bottled water}) 1
49 0.060349   frozenset({sausage}) 1
15 0.053131   frozenset({citrus fruit}) 1
41 0.051728   frozenset({pastry}) 1
42 0.049054   frozenset({pip fruit}) 1
50 0.047584   frozenset({shopping bags}) 1
10 0.046916   frozenset({canned beer}) 1
4 0.045312    frozenset({bottled beer}) 1
59 0.043708   frozenset({whipped/sour cream}) 1
37 0.038896   frozenset({newspapers}) 1
21 0.037760   frozenset({frankfurter}) 1
6 0.037626    frozenset({brown bread}) 1
43 0.037091   frozenset({pork}) 1

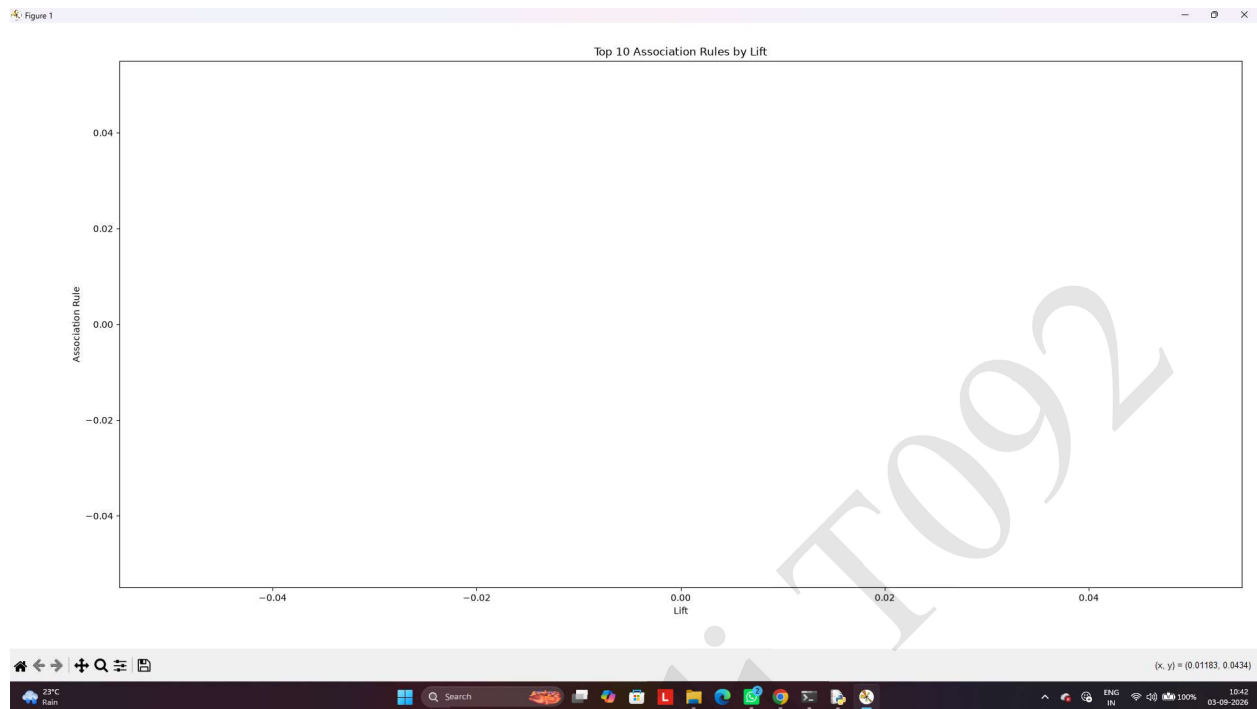
Association Rules:
Empty DataFrame
Columns: [antecedents, consequents, support, confidence, lift]
Index: []
```

**Graphs :-**



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**Observation :-**

**1. Top 10 Frequent Itemsets:**

The graph shows that some grocery items/item combinations occur more frequently than others. Itemsets with higher support are the most commonly purchased products.

**2. Top 10 Association Rules by Lift:**

The graph shows the strength of relationships between products. Rules with higher lift indicate stronger associations, meaning the products are more likely to be purchased together than independently.

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